



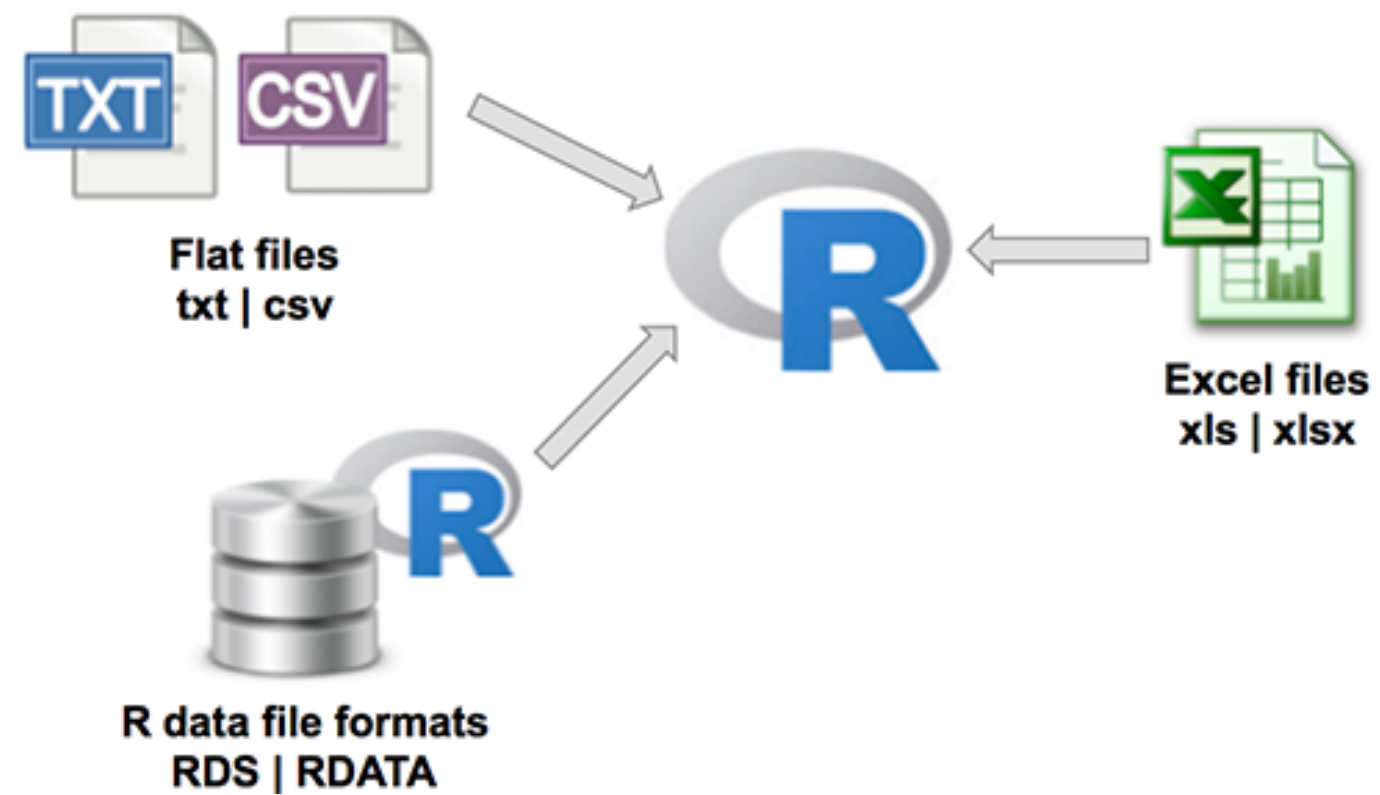
CANCER EPIGENETICS COURSE

DATA MANIPULATION

Presenter: Tran Duc Phuc

January 4th 2026

Data manipulation



Import/Export

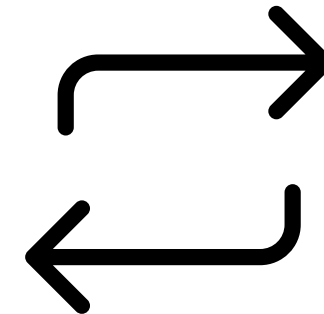


tidyr and dplyr

Wide data

Guest	Water	Wine	Soda
Alice	Yes	No	Yes
Bob	No	Yes	No

pivot_longer



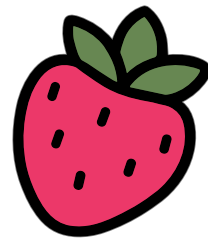
pivot_wider

Long data

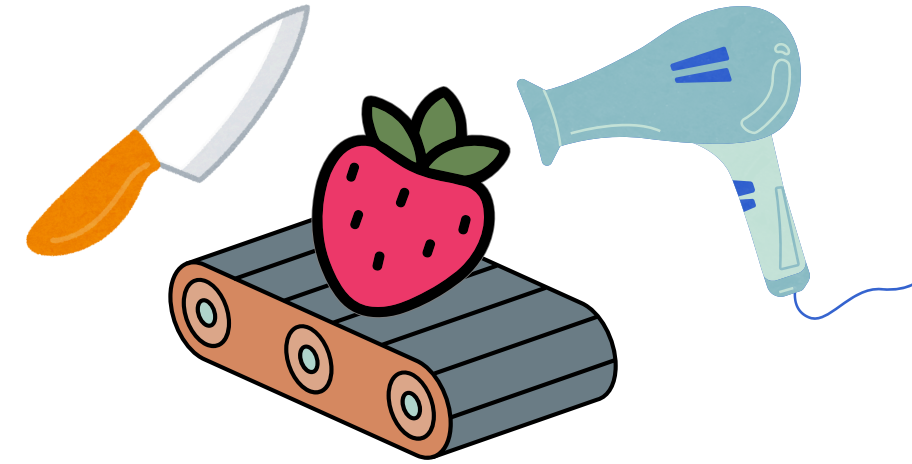
Guest	Drink	Ordered
Alice	Water	Yes
Alice	Wine	No
Alice	Soda	Yes
Bob	Water	No
Bob	Wine	Yes
Bob	Soda	No

- **Wide data:** suitable for humans (Excel,...). Each row represents a single person (or "subject"), and all their info is spread out horizontally.
- **Long data:** suitable for programming languages (especially ggplot2)

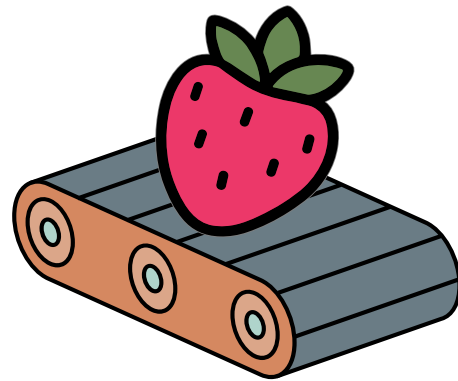
Think data as Raw material



**Raw material
(data)**



**Material processing
(dplyr)**



**Transport and reshape
by conveyor (tidyr)**



**Decorate material
(ggplot2)**